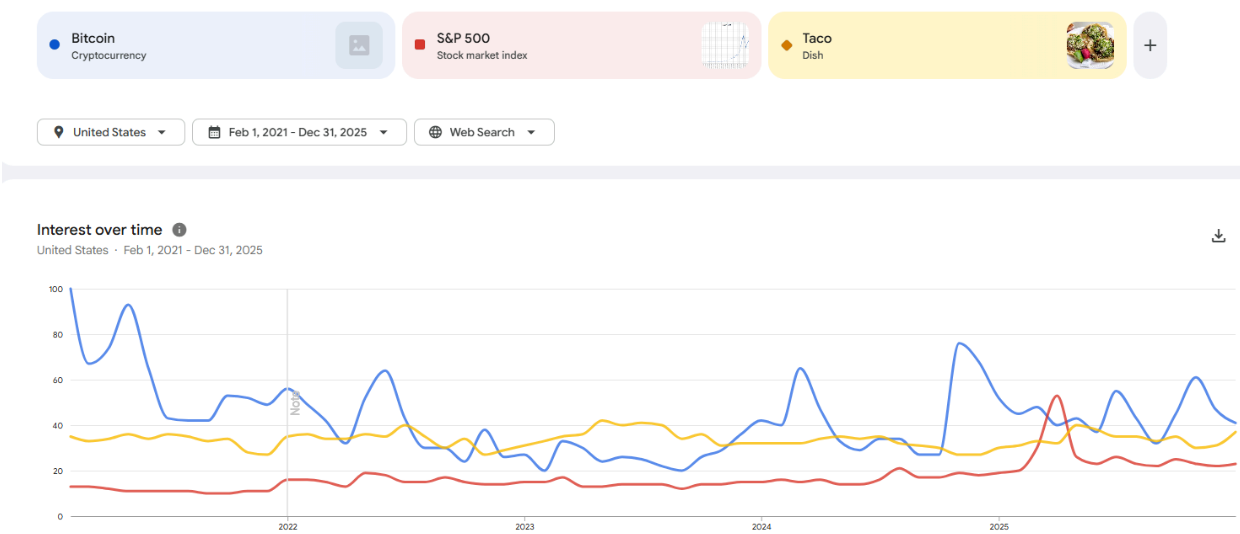
1. List of your Project Team Members (and a Team Name, if you'd like)
2. Tentative Project Title
3. Your 3 keyword/topic selections and a short description of why your team chose these topics.
4. The time period your data covers (and an explanation if you are using something other than the default 5 year period 1/1/2021 - 12/31/2025).
5. The location your search data is based on (and a short description of why it is the right location for your topics).
6. Plots (done in R) of each of the 3 keyword/topic time series shown individually and also, a separate plot showing all 3 together in the same plot. Plots should be labelled so that it is clear which series is associated with which keyword/topic.

1. Team Members : Ducheng Tan, Alexander Beckford

2. Project Title: *Trading Tuesday or Taco Tuesday?*

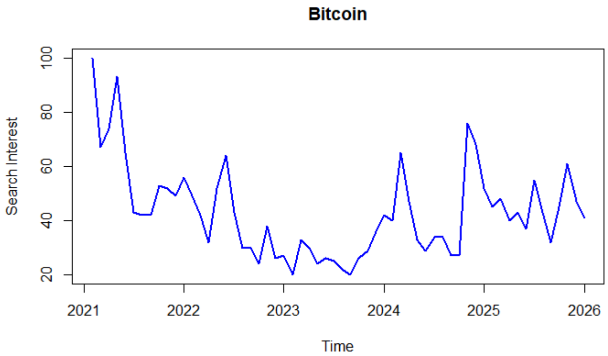
3. Topic Selections: Bitcoin (Cryptocurrency), S&P 500 (Stock Market Index), Taco (Dish).

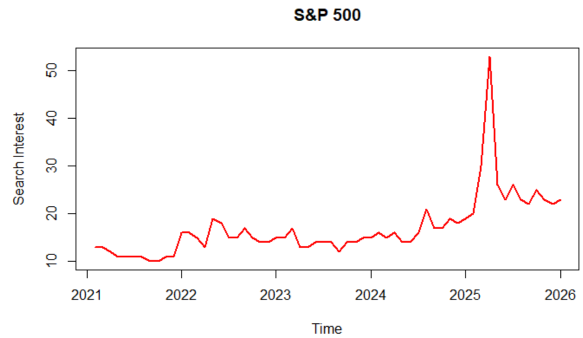
4. We use the default 5 year period 1/1/2021 - 12/31/2025 as suggested

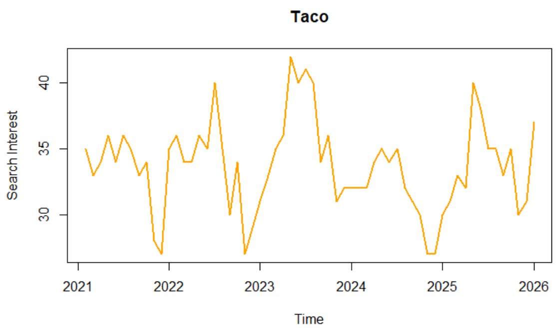


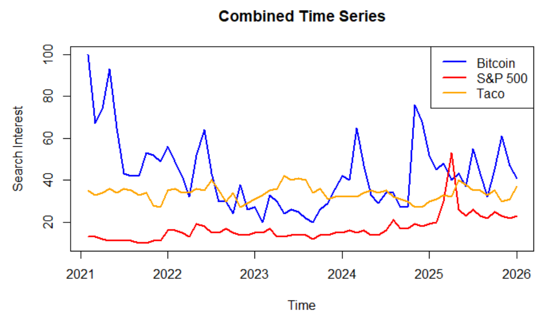
5. (Alex answer this)

6. Plots


Bitcoin
Search Interest
20
40
60
80
100
2021
2022
2023
2024
2025
2026
Time


S&P 500
Search Interest
10
20
30
40
50
2021
2022
2023
2024
2025
2026
Time


Taco
Search Interest
30
35
40
2021
2022
2023
2024
2025
2026
Time


Combined Time Series
Bitcoin
S&P 500
Taco
Search Interest
20
40
60
80
100
2021
2022
2023
2024
2025
2026
Time

Code:

```{r}

data <- read.csv("C:/Users/Ducheng Tan/R-Studio-Sync/STAT 335 Time Series/time_series_US_20210201-0000_20260210-1557.csv")

names(data)[names(data) == "S.P.500"] <- "SP500"
bitcoin_ts <- ts(data$Bitcoin, start = c(2021, 2), frequency = 12)
sp500_ts <- ts(data$SP500, start = c(2021, 2), frequency = 12)
taco_ts <- ts(data$Taco, start = c(2021, 2), frequency = 12)


plot.ts(bitcoin_ts, main = "Bitcoin", col = "blue",ylab="Search Interest", lwd = 2)
plot.ts(sp500_ts, main = "S&P 500", col = "red", ylab="Search Interest", lwd = 2)
plot.ts(taco_ts, main = "Taco", col = "orange",ylab="Search Interest", lwd = 2)


ts.plot(bitcoin_ts, sp500_ts, taco_ts,
        col = c("blue", "red", "orange"),
        lwd = 2,
        main = "Combined Time Series",
        ylab = "Search Interest")
legend("topright",
       legend = c("Bitcoin", "S&P 500", "Taco"),
       col = c("blue", "red", "orange"),
       lty = 1, lwd = 2)
```